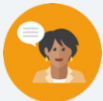


7.2 Defining Sine and Cosine

Lesson Introduction

 Benchmarks	MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles.		 Learning Target	I can define sine and cosine using similar triangles.
	MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.			
 Benchmark Coherence	Building On: MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.		Working Toward: N/A	
 Essential Questions	- How do I identify the opposite and adjacent sides in a right triangle? - What is a reference angle? - What are trigonometric ratios?			
 Lesson Narrative	Through exploration, students will determine that the ratio of the rise and run of similar right triangles will always be the same. Students will work together to discover the ratios involving the opposite, adjacent, and hypotenuse sides of right triangles, being formally introduced to the sine, cosine, and tangent of given reference angles.			
 Component Summary	7.2.1 Warm-Up (~5 min.)	Determine which region is a larger area (<i>SE p. 375</i>)	7.2.4 Bring It Together (5 min.)	Formalize the trigonometric ratios (<i>SE p. 379</i>)
	7.2.2 Exploration (10-15 min.)	Determine the relationships of slopes in similar right triangles (<i>SE p. 375</i>)	7.2.5 Guided Practice (5 min.)	Write trigonometric ratios given a reference angle (<i>SE p. 380</i>)
	7.2.3 Collaboration (10-15 min.)	Label the sides of a right triangle (<i>SE p. 378</i>)	7.2.6 Wrap-Up (~5 min.)	Write a letter to a friend describing the day's lesson (<i>SE p. 381</i>)
 Resources	Glossary		Materials	Additional Preparation
	Key Terms: - Sine - Cosine - Tangent Additional terms: - Opposite leg - Adjacent leg - Hypotenuse - Slope		- Straightedge - Protractor (Optional) Colored pencils (Optional) Tape	- This is a vocabulary-rich lesson. Consider creating several right triangles taped on the floor and/or wall for students to explore the vocabulary. - Consider taping a large right triangle on the floor of the classroom prior to the lesson for kinesthetic learners.

 Teacher Tips	Common Misconceptions	Things to Consider
	- During 7.2.3 Collaboration, students may believe only two of the angles are congruent. - Beginning in 7.2.3 Collaboration and throughout the lesson, students may mislabel the sides based on the reference angle.	- Make sure students are exposed to several different orientations of a right triangle, so that they are not only considering right triangles positioned in the same direction. - Use colored pencils to color code the sides when labeling triangles based on the reference angle.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Collect and Display During 7.2.2 Exploration, listen for and collect responses about the slopes of special right triangles. Facilitate a discussion with students about the different conjectures.	MA.K12.MTR.5.1: Patterns and Structure Throughout the lesson, students will be working to find patterns in the relationships. Students will first build relationships from patterns for 30-60-90 and 45-45-90 triangles. Then students will work with reference angles and ratios to discover the relationships between the ratios for sine, cosine, and tangent.
 Differentiate Instruction	An alternative way of reviewing content and working through 7.2.2 Exploration would be to have students work through 3 or 4 stations. - Station 1: On Ramp - Determine slope from a graph. - Station 2: 30-60-90 triangles - Students complete the first part of the exploration task. - Station 3: 45-45-90 triangles - Students complete the second part of the exploration task. (Optional) Station 4: Remediation or re-teaching the previous lesson based on data collected. Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

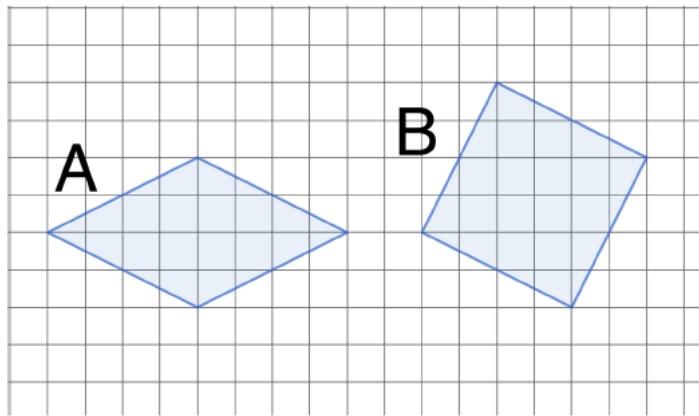
7.2.1 Warm-Up: Which Region Is Larger?

Component Overview: Students are determining which shaded region has a larger area and explaining their reasoning.



7.2.1 Warm-Up: Which Region Is Larger?

1. Two grids are shown with regions shaded.



- a. Which shaded region is larger?
Figure B has the larger shaded region.
- b. Explain your reasoning.
Answers will vary.



Watch to see how students reason about which area is larger. Do they try to use diagonal distances as equivalent to one gridline? Do they imagine decomposing the shaded regions to determine the number of whole squares?

7.2.2 Exploration: Similar Right Triangles

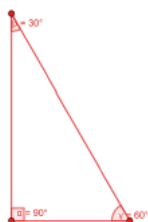
Component Overview: Students are exploring the slopes of the hypotenuses in 30-60-90 and 45-45-90 triangles.



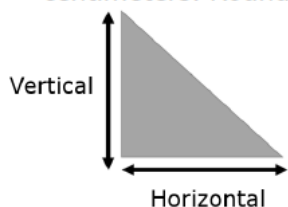
7.2.2 Exploration: Similar Right Triangles

- Using the space below, create a right triangle that has one angle with a measure of 60° . Be accurate with measurements.

Answers will vary.



- Measure the horizontal and vertical distance of your triangle. Determine the slope of your triangle in centimeters. Round to the nearest tenth.



Vertical	Horizontal	Slope ($\frac{\text{vertical}}{\text{horizontal}}$)
7.5	4.4	$\frac{7.5}{4.4} \approx 1.7$

Answers will vary.

- Record the slopes from two of your classmates. *You are the only person who should write in the first two columns of the table.* Have the person initial stating that your slope is correct.

Answers will vary.

Classmate	Slope	Initials

- What do you notice about the slopes?

Answers will vary. They are all approximately equal to 1.7.



An alternative way of reviewing content and working through the exploration activity would be to have students work through 3 stations:

Station 1: On Ramp - Determine Slope from a Graph

Station 2: 30-60-90 - Students complete the first part of the Exploration task.

Station 3: 45-45-90 - Students complete the second part of the Exploration task.

A 4th station could be utilized for remediation or re-teaching of the previous day's lesson.



Have students create a variety of different triangles using GeoGebra as they complete the exploration.

7.2.2 Exploration: Similar Right Triangles (continued)



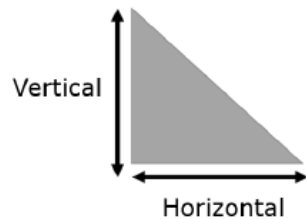
7.2.2 Exploration: Similar Right Triangles

- Using the space below, create a right triangle that has one angle with a measure of 60° . Be accurate with measurements.

Answers will vary.



- Measure the horizontal and vertical distance of your triangle. Determine the slope of your triangle in centimeters. Round to the nearest tenth.



Vertical	Horizontal	Slope ($\frac{\text{vertical}}{\text{horizontal}}$)
7.5	4.4	$\frac{7.5}{4.4} \approx 1.7$

Answers will vary.

- Record the slopes from two of your classmates. You are the only person who should write in the first two columns of the table. Have the person initial stating that your slope is correct.

Answers will vary.

Classmate	Slope	Initials

- What do you notice about the slopes?

Answers will vary. They are all approximately equal to 1.7.



How are the slopes similar if the 30- and 60-degree angles swap locations? How are they different?



Consider allowing students to write their notices and wonderings on two sticky notes and place them together in an area of the classroom for everyone to view. Another alternative is to use an online whiteboard service or forum space for students to post their responses.

7.2.2 Exploration: Similar Right Triangles (continued)

- a. What do you notice about the slopes?

Answers will vary. They are all approximately equal to 1.

- b. What do you wonder?

Answers will vary. Is this true for all triangles with a base of 45° ?

7. Ask two classmates for the "Notice and Wonder" what they wrote for both triangles. Record their responses below. *You are the only person who should write in the first three columns of the table.* Have the person initial that it is written correctly.

Answers will vary.

Classmate	Notice	Wonder	Initials
Classmate #1	<i>The right triangles we constructed with a base of 60° have the same approximate slope of 1.7.</i>	<i>Is this true for all right triangles with a base of 60°?</i>	
Classmate #2	<i>The right triangles we constructed with a base of 45° all have the same approximate slope of 1.0.</i>	<i>Is this true of all right triangles with a base of 45°?</i>	

8. Write a conjecture for each of the right triangles that were drawn.

Angle Measures	Conjecture
$30^\circ - 60^\circ - 90^\circ$	<i>Answers will vary. The slope of a $30^\circ - 60^\circ - 90^\circ$ triangle is approximately 1.7.</i>
$45^\circ - 45^\circ - 90^\circ$	<i>Answers will vary. The slope of a $45^\circ - 45^\circ - 90^\circ$ triangle is equal to 1.</i>



Does it matter which angle is used when finding the slope? Justify your response.



In future lessons, students will be working with 30° - 60° - 90° and 45° - 45° - 90° triangles. Without giving students the special right triangle rules, allow them to discuss anything they may have noticed or wondered about the triangles as they were completing the exploration activity.

7.2.2 Exploration: Similar Right Triangles (continued)

9. The triangles drawn by you and your classmates during the Exploration vary. Using either the triangles explored in Questions 1-3 or Questions 4-6, explain why these triangles are similar.

Each of the triangles that were drawn had two congruent angles, so all of the triangles explored in Questions 1-3 are similar by AA similarity and all of the triangles explored in Questions 4-6 are similar by AA similarity.



Right triangles with one other congruent angle are similar because of AA similarity.



Collect and Display

Listen for and collect responses about the slopes of special right triangles. Facilitate a discussion with students about the different conjectures.

7.2.3 Collaboration: Introduction to Trigonometric Ratios

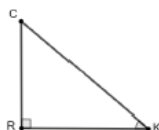
Component Overview: Students are discovering how to label the opposite and adjacent sides based on the reference angle. Students will then explore the relationship between sine, cosine, and tangent in similar triangles.



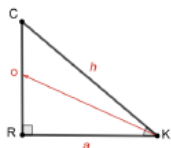
7.2.3 Collaboration: Introduction to Trigonometric Ratios

Work with your partner to complete the following.

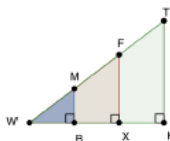
1. $\triangle CRK$ is shown, where $m\angle CRK = 90^\circ$ and $\angle CKR$ is marked as a reference angle.



- a. Draw an arrow from $\angle CKR$ to side $\overline{CR}$. Label that side with an o for opposite.
- b. Label side $\overline{RK}$ with an a for adjacent.
- c. Label the hypotenuse, $\overline{CK}$, with an h for hypotenuse.



2. The diagram shows three right triangles that share the reference $\angle W$. Explain why $\triangle WMB$, $\triangle WFX$, and $\triangle WTK$ are similar.



Answers will vary.

All three triangles have a right angle and all three share $\angle W$, therefore they are all similar by AA similarity postulate, which states that two or more triangles are similar if two angles in one triangle are congruent to two angles in another triangle.



For a kinesthetic approach for the lesson, allow students to use colored pencils to color code each side. Another option is to tape a right triangle to the floor, and have students walk to the opposite leg and reach down to touch the adjacent leg.



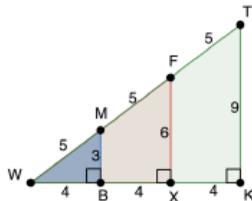
- Which side would be the opposite if $\angle KCR$ was the reference angle?
- Which side would be the adjacent side?
- Why would the sides change if we used a different reference angle?

7.2.3 Collaboration: Introduction to Trigonometric Ratios (continued)

3. Complete the table.

Triangle	Leg Opposite to $\angle W$	Leg Adjacent to $\angle W$	Hypotenuse
$\triangle WMB$	$\overline{MB}$	$\overline{WB}$	$\overline{WM}$
$\triangle WFX$	$\overline{FX}$	$\overline{WX}$	$\overline{WF}$
$\triangle WTK$	$\overline{TK}$	$\overline{WK}$	$\overline{WT}$

4. The diagram shows measurements for the three triangles.



Complete the table using angle $\angle W$ as the reference angle.

Triangle	$\frac{\text{opposite}}{\text{hypotenuse}}$	$\frac{\text{adjacent}}{\text{hypotenuse}}$	$\frac{\text{opposite}}{\text{adjacent}}$
$\triangle WMB$	$\frac{3}{5}$	$\frac{4}{5}$	$\frac{3}{4}$
$\triangle WFX$	$\frac{6}{10} = \frac{3}{5}$	$\frac{8}{10} = \frac{4}{5}$	$\frac{6}{8} = \frac{3}{4}$
$\triangle WTK$	$\frac{9}{15} = \frac{3}{5}$	$\frac{12}{15} = \frac{4}{5}$	$\frac{9}{12} = \frac{3}{4}$



Common Misconception

Students may believe only two of the angles are congruent. Facilitate a discussion about how the sum of a triangle is 180, and if 2 angles are congruent, then the 3rd angle has to be congruent.



For a visual and kinesthetic approach, consider having students trace the three triangles on patty paper. Students can label side lengths and mark congruent sides.

7.2.4 Bring It Together: Trigonometric Ratios

Component Overview: Using the information from 7.2.2 Exploration and 7.2.3 Collaboration, students will bring together all their findings to define sine, cosine, and tangent.



7.2.4 Bring It Together: Trigonometric Ratios

Since $\triangle WMB$, $\triangle WFX$, and $\triangle WTK$ are similar, the $\frac{\text{opposite}}{\text{hypotenuse}}$ ratios are equivalent to each other. This is also true for the ratios $\frac{\text{adjacent}}{\text{hypotenuse}}$ and $\frac{\text{opposite}}{\text{adjacent}}$. For this reason, it is possible to define these ratios in any right triangle.

The ratios use the reference angle theta (θ).



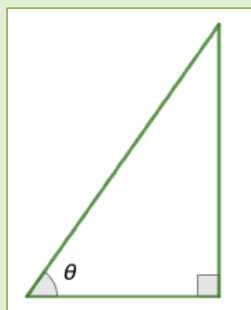
Facilitate a discussion over the importance of knowing the reference angle when using sine, cosine, and tangent.

Trigonometric Ratios

$$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}}$$



7.2.5 Guided Practice: Trigonometric Ratios

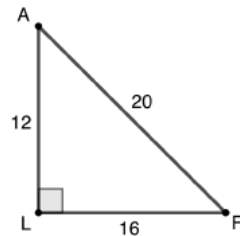
Component Overview: Students are determining the ratio of sides in right triangles when given a reference angle.



7.2.5 Guided Practice: Trigonometric Ratios

1. $\triangle FLA$ is shown. $m\angle ALF = 90^\circ$, $AL = 12$, $LF = 16$, and $AF = 20$.

Complete the table using each angle as the reference angle.

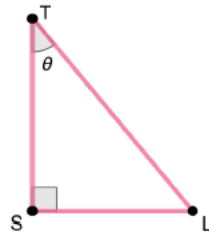


Angle	$\frac{\text{opposite}}{\text{hypotenuse}}$	$\frac{\text{adjacent}}{\text{hypotenuse}}$	$\frac{\text{opposite}}{\text{adjacent}}$
$\angle F$	$\frac{12}{20} = \frac{3}{5}$	$\frac{16}{20} = \frac{4}{5}$	$\frac{12}{16} = \frac{3}{4}$
$\angle A$	$\frac{16}{20} = \frac{4}{5}$	$\frac{12}{20} = \frac{3}{5}$	$\frac{16}{12} = \frac{4}{3}$

2. $\triangle TLS$ is shown where $LS = \sqrt{35}$ and $TS = \sqrt{42}$.
a. Determine the length of TL .

$$(\sqrt{35})^2 + (\sqrt{42})^2 = (TL)^2$$

$$TL = \sqrt{77}$$



- b. Complete the table for the reference angle θ .

Angle	$\frac{\text{opposite}}{\text{hypotenuse}}$	$\frac{\text{adjacent}}{\text{hypotenuse}}$	$\frac{\text{opposite}}{\text{adjacent}}$
θ	$\frac{\sqrt{35}}{\sqrt{77}}$	$\frac{\sqrt{42}}{\sqrt{77}}$	$\frac{\sqrt{35}}{\sqrt{42}}$



Common Misconception

Students may mislabel the sides based on the reference angle. Allow students to use a colored pencil to color code the sides, or to redraw the triangle to find the sides for the second reference angle.



Common Misconception

If students determine the length is 77, then they did not take the square root of c^2 . Facilitate a discussion over inverse operations in order to discuss how to find c .



Consider challenging students to discover the relationship between the ratios of sine and cosine to the ratio of tangent.

- How can you use the ratios for sine and cosine to find the ratio for tangent?



7.2.6 Wrap-Up: Cool Down

Component Overview: Students complete a letter to a friend explaining the day's lesson.



7.2.6 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to explain what was missed in today's lesson.

Answers will vary.



What would a note look like for you as a teacher? Consider your own letter and the vocabulary you would expect to see in the letter. Student responses will also give you insight to the level of understanding of the lesson.